

Gauss gun — Solution

X25 (2025) — English translation

*

-2.00 pts

Attention! If your steel ball rolls under someone else's table, then it is forbidden to take it out! Spare balls and magnets are not issued! Marking ANY kind of mark on an aluminum gutter will result in a penalty point. Losing ANY equipment other than balls or a magnet will result in a penalty point. Intentionally changing the angle of an aluminum gutter will disqualify you.

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A1

0.10 pts

Specify the value h . Determine the installation angle α .

Let's measure the height of the lower horizontal part of the gutter $h = 24.9 \text{ cm}$. Let's measure the height of the top point $h + \Delta h = 50.7 \text{ cm}$. In this case, the length of the long part of the gutter is $L = 80.0 \text{ cm}$. Using the fact that $\sin \alpha = \Delta h / L$ we find the angle α

Answer:

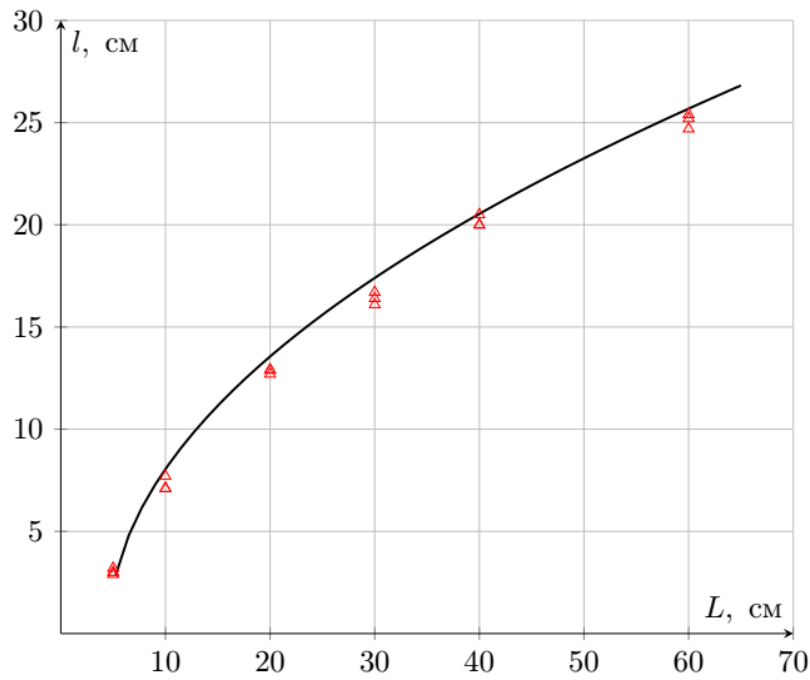
$$h = 24.9 \text{ cm} \quad \alpha = 18.8^\circ$$

A2

1.00 pts

Measure the dependence of the flight range l on the length of the inclined section of the ball path L in the range from 0 to 70 cm for 5 values L . Repeat each measurement at least 3 times. All measurements l must be taken using ball drop marks on A3 sheets. A3 sheets are required to be handed in at the end of the work.

Answer: L , cm l_1 , cm l_2 , cm l_3 , cm $\langle l \rangle$, cm 10.0 7.1 7.1 7.7 7.8 20.0 12.7 12.9 12.9 13.3 30.0 16.1 16.4 16.7 16.9 40.0 20.0 20.0 20.5 20.7 60.0 24.7 25.2 25.4 25.6



A3

0.40 pts

Obtain a theoretical expression relating the quantities l and L . Assume that the ball moves along the chute all the time without slipping; the angle between the generatrices of the chute is straight. Express your answer using h, α, γ, l, L

Kinetic energy of the ball taking into account the rotational energy:

$$K = \frac{Mv^2}{2} + \frac{I\omega^2}{2},$$

where $I = \frac{2}{5}MR^2$. The speed of the ball points that touch the corner must be zero, so $\omega R/\sqrt{2} = v$. As a result, on the inclined section of the chute, the ESE has the form

$$K = \Delta\Pi \Rightarrow \frac{9Mv^2}{10} = MgL \sin \alpha$$

After passing the corner of the chute, γK of energy is lost, so the ball flies out of the corner with a horizontal speed u such that

$$u^2 = \frac{10}{9}(1 - \gamma)gL \sin \alpha$$

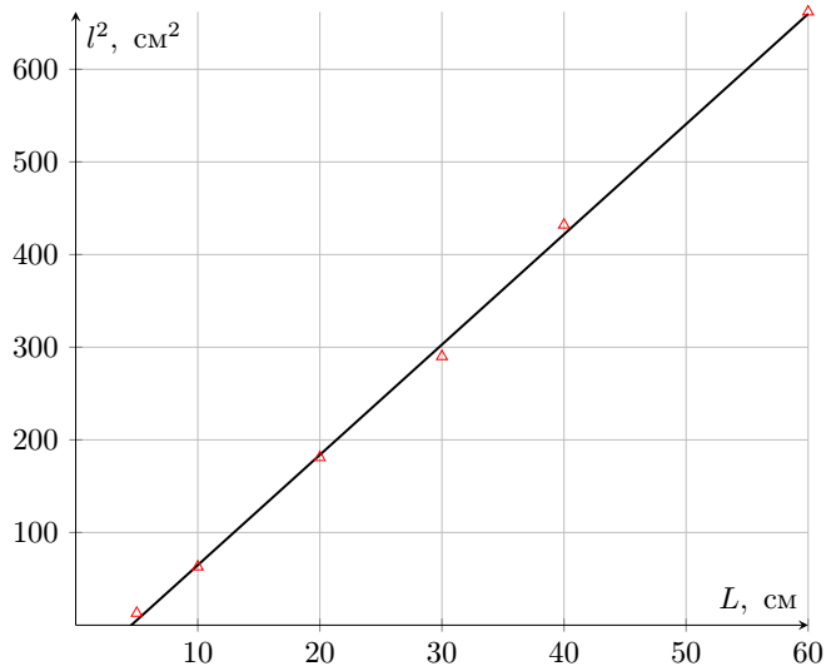
The ball, under the influence of gravity acceleration, overcomes the height h in time $t = \sqrt{2h/g}$, and therefore

$$l^2 = u^2 t^2 = \frac{20}{9}(1 - \gamma)Lh \sin \alpha$$

A4

0.60 pts

Using the result of the previous paragraph, construct a graph of l versus L in coordinates in which it will be linear.



A5

0.20 pts

Find the slope coefficient of the graph constructed in the previous paragraph. Determine the loss factor γ .

The graph in axes l^2 from L turns out to be linear with a slope coefficient

$$k_{A4} = \frac{20}{9}(1 - \gamma)h \sin \alpha = 11.9 \text{ cm}$$

The coefficient γ can vary quite a lot from corner to corner.

Answer:

$$\gamma = 1 - \frac{9k_{A4}}{20h \sin \alpha} = 0.36$$

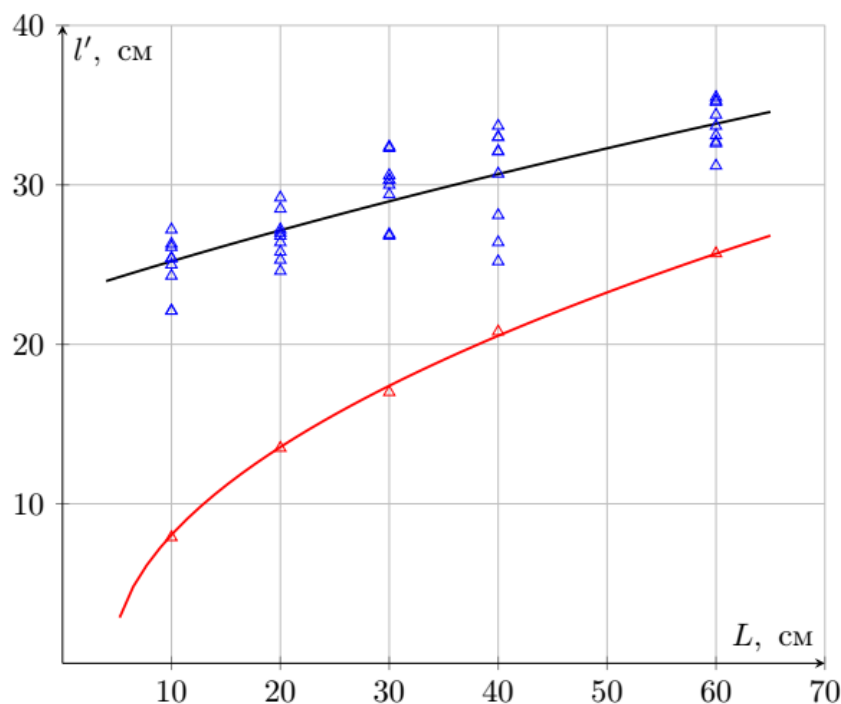
B1

2.00 pts

Measure the dependence of the flight range l' of the emitted ball on the length of the inclined section of the path of the incoming ball L . Take measurements for the same values of quantity L that you used in Part A. For each value of L , take at least 5 measurements of quantity l' . All measurements l must be taken using marks from the fall of the ball on A3 sheets.

All measurements l must be taken using marks from the fall of the ball on A3 sheets.

Answer: L , cm $\langle l' \rangle$, cm 10.0 25.4 20.0 26.8 30.0 30.3 40.0 30.5 60.0 34.0



B2

0.30 pts

Obtain an expression for the dependence $E_{kin}(l)$.

From the kinematics described above, we remove the rotational motion

$$E_{kin} = \frac{Mu^2}{2} = \frac{Mgl^2}{4h}$$

Answer:

$$E_{kin} = \frac{Mgl^2}{4h}$$

B3

0.40 pts

Using the results of steps A2 and B1, calculate the addition ΔE_{kin} to the kinetic energy of the projectile translational motion obtained during interaction with the gun, for each experimental value. Calculate the energy using the formula obtained in the previous paragraph, discarding constant factors in front of the function of l .

$$\Delta E_{kin} = l'^2 - l^2$$

Answer: $L, \text{ cm}$ $\Delta E_{kin}, \text{ cm}^2$ $\Delta E_{kin}, \text{ mJ}$ 10.0 584 2.36 20.0 534 2.16 30.0 627 2.53 40.0 497 2.00 60.0 492 1.99

B4

0.40 pts

Plot a graph of ΔE_{kin} versus l in coordinates in which it is linear.

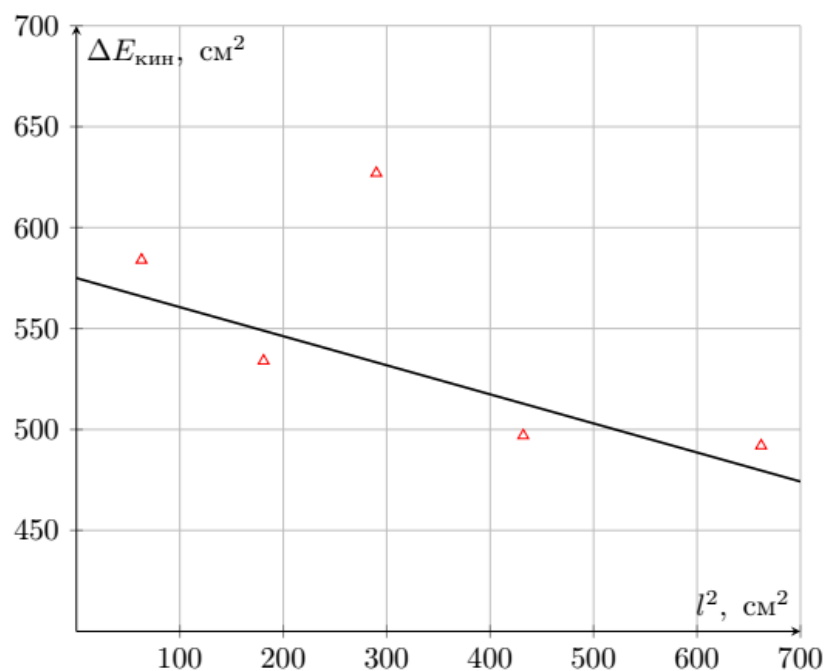
Energy ratio:

$$E'_{kin} = (E_{kin} + W_1)(1 - \beta) + W_2,$$

therefore

$$\Delta E_{kin} = -\beta l^2 + W_1(1 - \beta) + W_2$$

and the graph must be plotted in coordinates ΔE_{kin} from l^2



B5

0.20 pts

Using the graph from point B4, determine the value of β , as well as the maximum possible addition to the kinetic energy ΔE_{max} observed in the experiment.

The slope of the graph is $k_{B4} = \beta$, and the intercept is ΔE_{max} .

Answer:

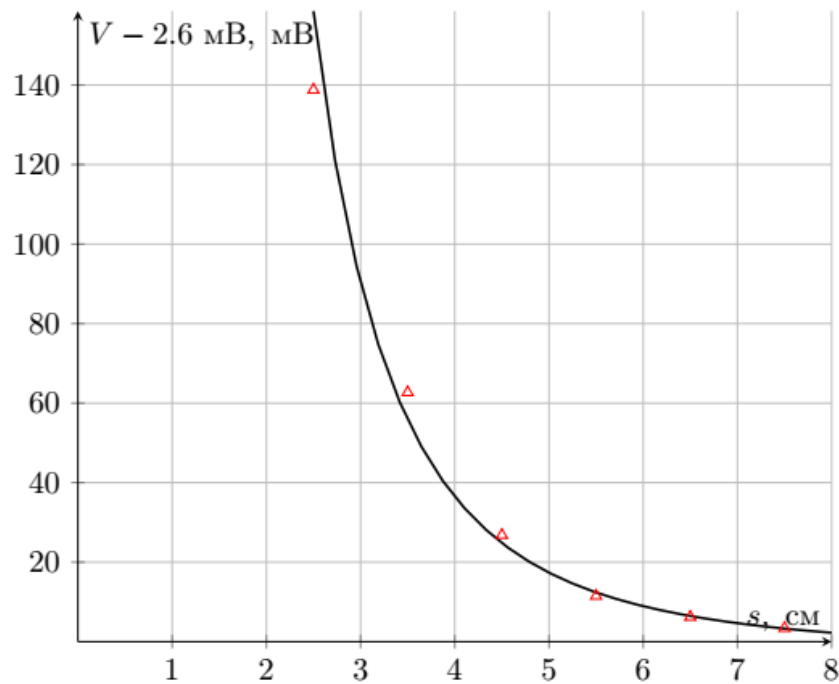
$$\beta = 0.15, \quad \Delta E_{\text{kin}} = 580 \text{ cm}^2 = 2.3 \text{ mJ}$$

C1

0.30 pts

Using a Hall sensor, measure the magnet field for 5 values of s .

Answer: $s, \text{ cm}$ 2, 6 mV, mV 2.5 138.8 3.5 62.7 4.5 26.8 5.5 11.5 6.5 6.2 7.5 3.4



C2

0.10 pts

Give the dimensions of m and M in SI units.

Answer:

$$[m] = A \cdot m^2$$

$$[M] = A \cdot m^{-1}$$

C3

0.30 pts

Obtain the theoretical form of the dependence $B(s)$. Express the answer through the quantities μ_0, m, s , where m is the magnetic moment of the issued ball magnet. Specify the direction of the magnetic field at the point where the sensor is located.

Magnetic dipole field:

$$\vec{B} = \frac{\mu_0}{4\pi} \left(\frac{3(\vec{m}\vec{r})\vec{r}}{r^5} - \frac{\vec{m}}{r^3} \right)$$

in projection on the axis directed along the dipole moment $\vec{m}$:

$$B(s) = \frac{\mu_0}{4\pi} \cdot \frac{2m}{s^3}$$

Answer:

$$B(s) = \frac{\mu_0}{4\pi} \cdot \frac{2m}{s^3}$$

Direction - along magnetic moment $\vec{m}$.

C4

0.60 pts

Using the results of the previous two paragraphs, construct a graph of the dependence $B(s)$ in coordinates in which it will be linear. Determine the magnetization value of the magnet M .

We build the graph in coordinates $B^{-1/3}$ from s , where B is calculated by the formula:

$$B = \frac{V}{31.25 \text{ V/T}}$$

Coordinate s is chosen without a degree because the data along it is distributed more evenly.

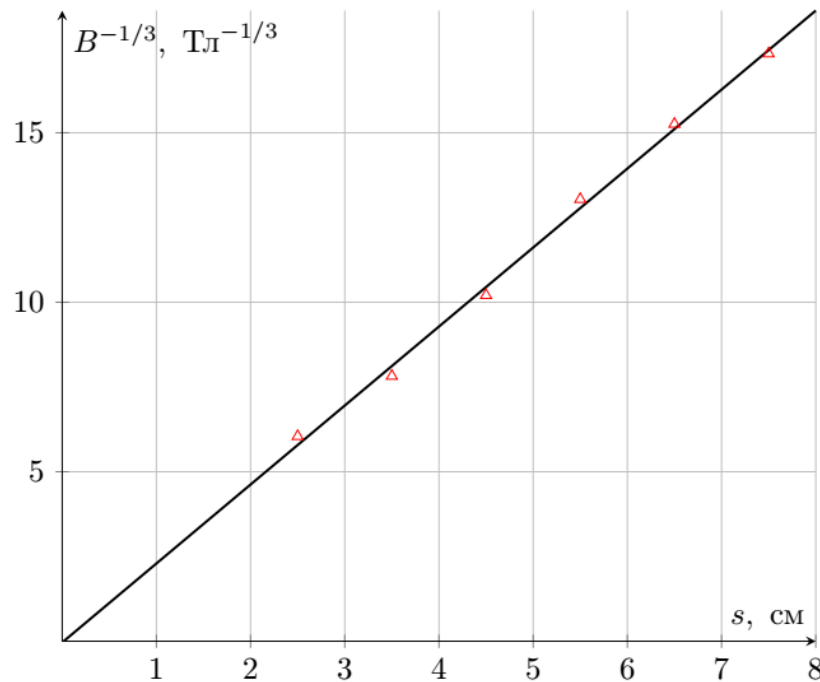
The slope coefficient of the graph is

$$k_{C4} = \left(\frac{2\pi}{m\mu_0} \right)^{-1/3} = 2.33 \frac{\text{T}^{-1/3}}{\text{cm}}$$

In this case, magnetization is the volumetric density of the magnetic moment, i.e.

$$M = \frac{3m}{4\pi R_0^3}, \quad m = \frac{2\pi}{k_{C4}^3 \mu_0} = 0.36 \text{ A} \cdot \text{m}^2$$

The magnetization value given in the condition is deliberately overestimated. N38 brand magnets issued in the work have a magnetization of $M \in [6.76, 7.03] \cdot 10^5 \text{ A/m}$

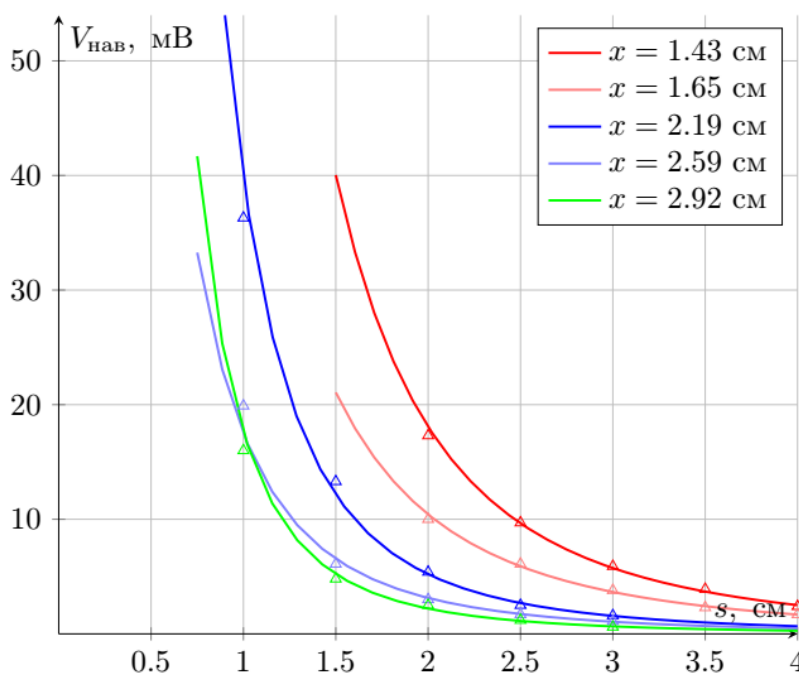


C5

1.50 pts

For each tube, measure the x value using a caliper. Also, for each tube, obtain the dependence $B_{ind}(s)$ by removing at least 5 experimental points.

Answer: x , cm 1.43 x , cm 1.65 $s - 0.5$ cm, cm $V_{tot} + 2.6$ mV, mV $V_{mag} + 2.6$ mV, mV $s - 0.5$ cm, cm $V_{tot} + 2.6$ mV, mV $V_{mag} + 2.6$ mV, mV 3.5 10.9 8.5 3.5 8.6 6.9 3.0 17.6 13.7 3.0 13.3 11.0 2.5 26.3 20.4 2.5 20.5 16.7 2.0 39.9 30.2 2.0 31.0 24.9 1.5 65.8 48.5 1.5 46.2 36.2 0.0 768 275 0.0 487 172.0 x , cm 2.2 x , cm 2.59 $s - 0.5$ cm, cm $V_{tot} + 2.6$ mV, mV $V_{mag} + 2.6$ mV, mV $s - 0.5$ cm, cm $V_{tot} + 2.6$ mV, mV $V_{mag} + 2.6$ mV, mV 2.5 12.8 11.2 2.5 8.8 7.7 2.0 17.8 15.3 2.0 13.3 11.6 1.5 30.1 24.7 1.5 19.4 16.4 1.0 52.4 39.1 1.0 30.7 24.6 0.5 97.8 61.5 0.5 59.0 39.1 0.0 264 100.0 0.0 167.4 64.8 x , cm 2.92 $s - 0.5$ cm, cm $V_{tot} + 2.6$ mV, mV $V_{mag} + 2.6$ mV, mV 2.5 6.8 6.2 2.0 9.9 8.7 1.5 15.8 13.3 1.0 23.0 18.2 0.5 45.8 29.8 0.0 113.0 44.4



C6

0.50 pts

For each tube, using the results of the previous paragraph, using the least squares method, determine the dipole moment m' induced in the steel ball. There is NO need to create graphs. Choose a linearization such that a shift in the position of the dipole induced in the steel ball (i.e., a constant shift s) does not affect the linearity.

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$$B(s) = \frac{\mu_0}{4\pi} \cdot \left(\frac{2m'}{s^3} + \frac{2m}{(s+x)^3} \right)$$

At the same time the induced field gives a linearization $V^{-1/3}$ from s .

Answer: x , cm m' , A · m² 1.43 0.030 1.65 0.021 2.19 0.0077 2.59 0.0057 2.92 0.0031

C7

0.40 pts

Write down the boundary conditions for the vectors $\vec{B}$ and $\vec{H}$ at the steel-air interface. Using the boundary conditions, obtain the relationship between the angles that form the magnetic field induction vector $\vec{B}$ with the normal to the surface in both media. The resulting relationship may contain the specified angles and μ_{steel} . Obtain an expression for the angle between the magnetic field induction vector and the normal in air in the limiting transition $\mu_{\text{steel}} \gg 1$.

1 - index of quantities in steel 2 - index of quantities in air As a consequence of the Gauss theorem for the Magnetic field, the normal component of the magnetic field $\vec{B}$ is preserved:

$$B_{n1} = B_{n2}$$

in the system there are no surface magnetic currents, so the tangential component $\vec{H}$ be conserved:

$$H_{\tau1} = H_{\tau2}$$

Use relation $\vec{B} = \mu\mu_0\vec{H}$:

$$B_{\tau1}/\mu = B_{\tau2}$$

$$B_{n1} \tan \theta_1 / \mu = B_{n2} \tan \theta_2$$

$$\tan \theta_1 / \mu = \tan \theta_2$$

For $\mu \gg 1$, obtain $\theta_2 = 0$.

Answer:

$$\begin{cases} B_{n1} = B_{n2} \\ H_{\tau 1} = H_{\tau 2} \end{cases}$$

$$\tan \theta_{\text{steel}} = \tan \theta_{\text{air}} \cdot \mu_{\text{steel}}$$

$$\theta_2 = 0$$

C8

0.10 pts

What type of substance does $\varepsilon \gg 1$ correspond to in electrostatics?

The boundary conditions in a conductor in electrostatics are similar before the replacement:

$$\vec{H} \Leftrightarrow \vec{E}$$

$$\vec{B} \Leftrightarrow \vec{D}$$

The limit $\varepsilon \rightarrow \infty$ in electrostatics is a transition to the conductor.

Answer: Conductor

C9

0.20 pts

At a distance $y > R$ from a ferromagnetic sphere of radius R there is a magnetic charge q_m . The external field created by a ferromagnetic sphere is equivalent to a system of magnetic charges. Describe the equivalent system of magnetic charges for this situation (find their magnitudes and positions). Note. Magnetic charge q_m creates a magnetic field $\vec{B}$ at a point with radius vector $\vec{r}$ according to the law:

$$\vec{B} = \frac{\mu_0 q_m}{4\pi r^3} \vec{r}$$

Describe the equivalent system of magnetic charges for this situation (find their magnitudes and positions).

Note. Magnetic charge q_m creates a magnetic field $\vec{B}$ at a point with radius vector $\vec{r}$ according to the law:

$$\vec{B} = \frac{\mu_0 q_m}{4\pi r^3} \vec{r}$$

By analogy with electrostatics, a charge-image will appear in the sphere at a distance R/y from the center and with a value $q_1 = -q_m \cdot R/y$. However, from Gauss's theorem for a magnetic field for a surface containing a sphere, it follows that the total magnetic charge is zero. Therefore, a charge $-q_1$ will be induced at the center of the sphere.

Answer: At the center of the sphere there is a magnetic charge $q_m \cdot R/y$ At a distance R/y from the center of the sphere $-q_m \cdot R/y$

C10

0.20 pts

Obtain an expression for the magnetic moment m' induced in a steel ball radius R when it is at a distance x from the magnetic dipole m .

Let x be the distance from the center of the sphere to the positive charge of the dipole $+q_m$. According to the previous paragraph, a dipole moment $m_1 = \frac{R^2}{x} \cdot q \frac{R}{x} = \frac{qR^3}{x^2}$ will be induced in the sphere. The second charge is located at a distance of $x + l$ from the center of the sphere, where l is the arm of the dipole m . The dipole moment induced by the second charge is: $m_2 = -\frac{qR^3}{(x+l)^2}$ Total dipole moment:

$$m' = m_1 + m_2 = qR^3\left(\frac{1}{x^2} - \frac{1}{(x+l)^2}\right) \approx qR^3\frac{2l}{x^3} = \frac{2mR^3}{x^3}$$

Answer:

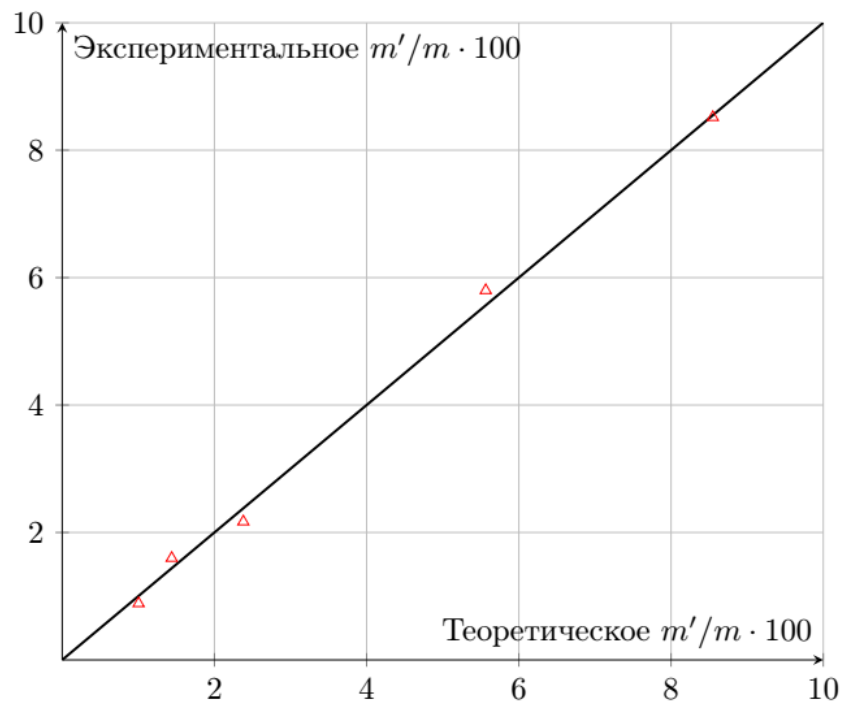
$$m' = \frac{2mR^3}{x^3}$$

C11

0.80 pts

Using the relationship obtained in the previous paragraph, linearize the measurement results in paragraph C6 and plot it. Find the parameters of the line and compare them with the theoretical values.

The graph can be plotted in coordinates m' from $1/x^3$. Below is a plot of the experimental value m'/m versus the theoretical value $2R^3/x^3$.



D1

0.20 pts

In a steel ball located at a distance $s_1 = 2R$ from a dipole with a dipole moment m , a dipole moment $t_1 m$ is induced, and at a distance $s_2 = 4R$ a moment $t_2 m$ is induced. Determine t_1 and t_2 . The steel balls do not affect the m dipole!

Magnetic field at a distance of $2R$ from the dipole:

$$B(2R) = \frac{\mu_0 m}{2\pi} \frac{1}{8R^3}$$

$$t_1 m = \frac{4\pi R^3}{\mu_0} \cdot B(2R) = \frac{m}{4}$$

$$t_1 = \frac{1}{4}$$

Magnetic field at a distance of $4R$ from the dipole:

$$B(4R) = \frac{\mu_0 m}{2\pi} \frac{1}{4^3 R^3}$$

$$t_2 m = \frac{4\pi R^3}{\mu_0} \cdot B(4R) = \frac{m}{32}$$

$$t_2 = \frac{1}{32}$$

Answer:

$$t_1 = 1/4$$

$$t_2 = 1/32$$

D2

0.60 pts

Obtain an expression for the induced dipole moments m_1, m_2 in the ball-magnet-ball system and m_3, m_4 in the ball-ball-magnet system. Express your answer using m . Note. Don't forget to take into account the influence of the balls on each other.

Let's create a system of equations describing the consistency of magnetic moments:

$$\begin{cases} m_1 t_2 + m t_1 = m_2 \\ m t_1 + m_2 t_2 = m_1 \end{cases}$$

$$\begin{cases} m t_2 + m_4 t_1 = m_3 \\ m t_1 + m_3 t_1 = m_4 \end{cases}$$

Answer:

$$m_1 = m_2 = 8/31 m$$

D3

0.30 pts

Obtain an expression for the magnetic field induction $B(x)$ of the ball-ball-magnet system at a distance x from the center of the magnet. Express your answer using m, μ_0, x, R .

Answer:

$$B(x) = \frac{\mu_0}{2\pi} \left(\frac{m}{x^3} + \frac{m_4}{(x+2R)^3} + \frac{m_3}{(x+4R)^3} \right)$$

D4

0.30 pts

Obtain an expression for the magnetic field induction $B(x)$ of the ball-magnet-ball system at a distance x from the center of the magnet. Express your answer using m, μ_0, x, R .

Answer:

$$B(x) = \frac{\mu_0}{2\pi} \left(\frac{m_2}{(x+2R)^3} + \frac{m}{x^3} + \frac{m_1}{(x-2R)^3} \right)$$

D5

0.30 pts

Obtain an expression for the work W_1 performed by the gun's magnetic field on the incoming ball in the form of the integral of the force acting on it. You will receive the coefficient in front of the functional part in numerical form. Express your answer using μ_0, R, m .

The force acting on the ball is $F = m \frac{dB}{dx}$. Moreover, $m \propto B$, therefore

$$W_1 = \int_{\infty}^{2R} \frac{4\pi R^3}{\mu_0} B \frac{dB}{dx} dx = \frac{2\pi R^3}{\mu_0} B^2(2R)$$

Answer:

$$W_1 = 2.68 \cdot 10^{-3} \frac{\mu_0 m^2}{R^3}$$

D6

0.30 pts

Obtain an expression for the work W_2 performed by the magnetic field of the gun on the flying ball in the form of the integral of the force acting on it. You will receive the coefficient in front of the functional part in numerical form. Express your answer using μ_0, R, m .

$$W_2 = \int_{-4R}^{-\infty} \frac{4\pi R^3}{\mu_0} B \frac{dB}{dx} dx = -\frac{2\pi R^3}{\mu_0} B^2(4R)$$

Answer:

$$W_2 = -3.83 \cdot 10^{-4} \cdot \frac{\mu_0 m^2}{R^3}$$

D7

0.40 pts

Using the expression for W_1 , W_2 and the values β , m , R , determine the theoretical value of $\Delta E_{\max}$.

Let's find the numerical values of $W_1 = 3.38$ mJ, $W_2 = -0.48$ mJ and calculate $E_{\max}$ using the formula

$$E_{\max} = W_1(1 - \beta) + W_2$$

It can also be noted that our estimate for $E_{\max}$ practically coincides with the measured value despite the fact that two uncontrolled approximations are used: The incoming ball does not change the magnetic moments of the gun balls. The flying ball also does not change the magnetic moments of the gun. From the point of view of the external magnetic field, all balls are point dipoles located in the center of the balls.

It can also be noted that our estimate for $E_{\max}$ is almost identical to the measured value despite the fact that two unsupervised approximations are used:

Answer:

$$E_{\max} = 2.38 \text{ mJ}$$